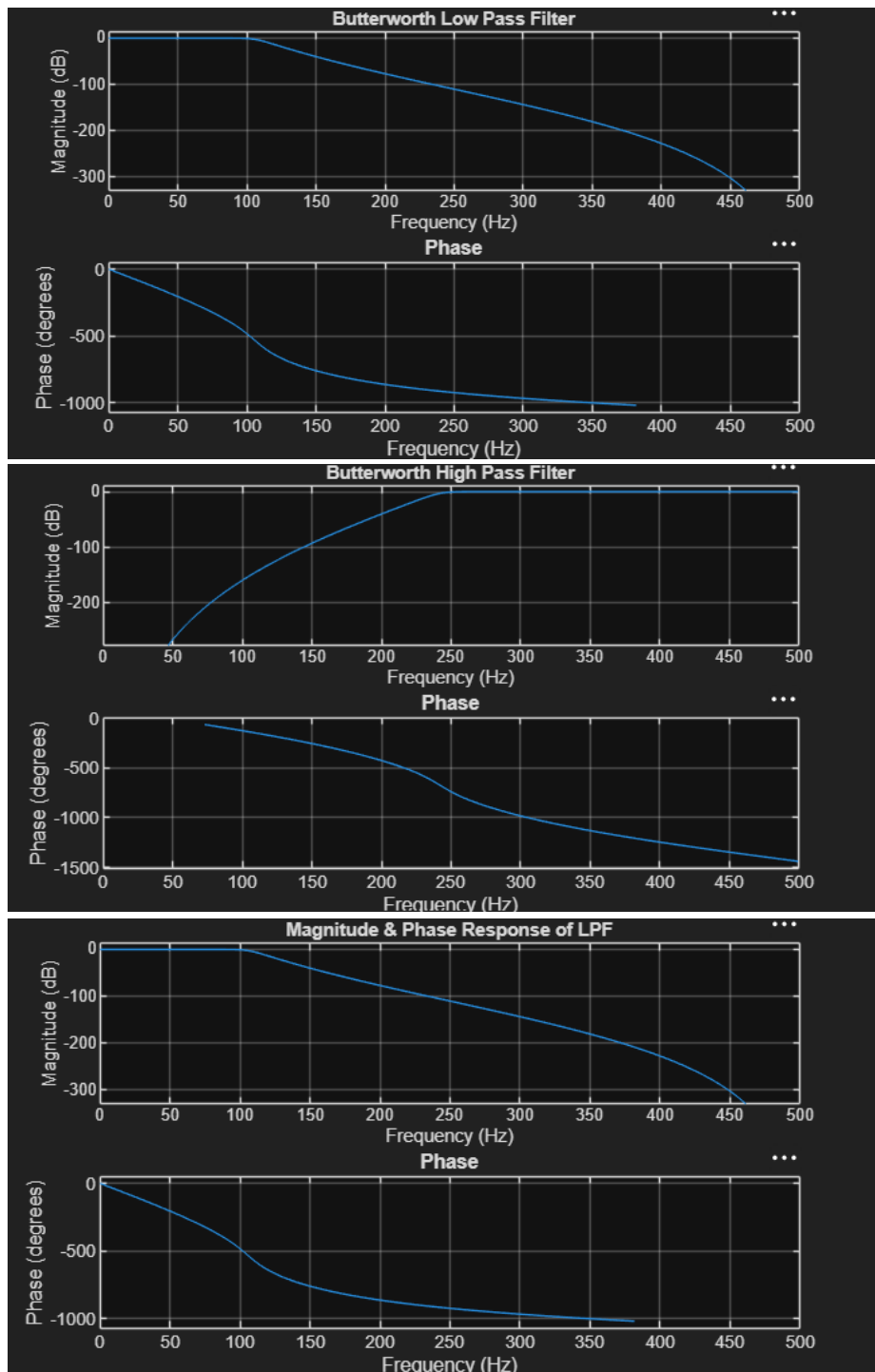
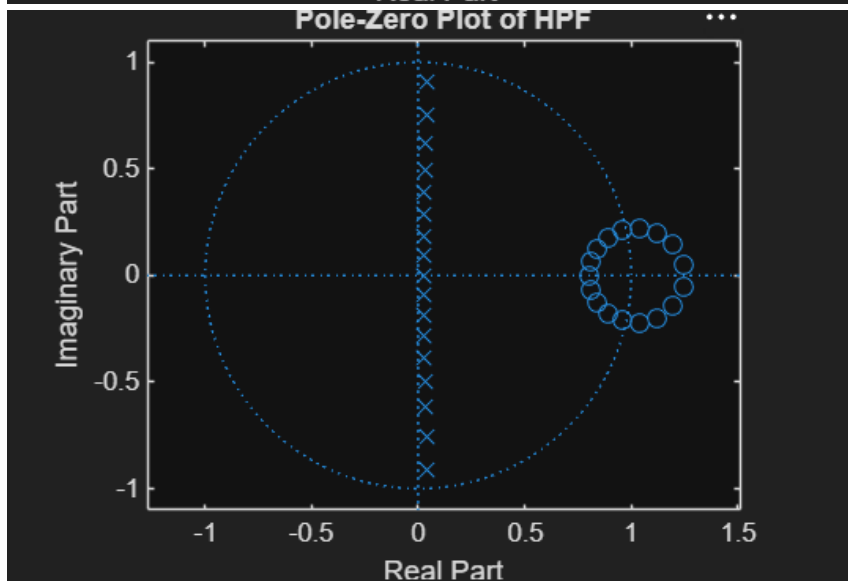
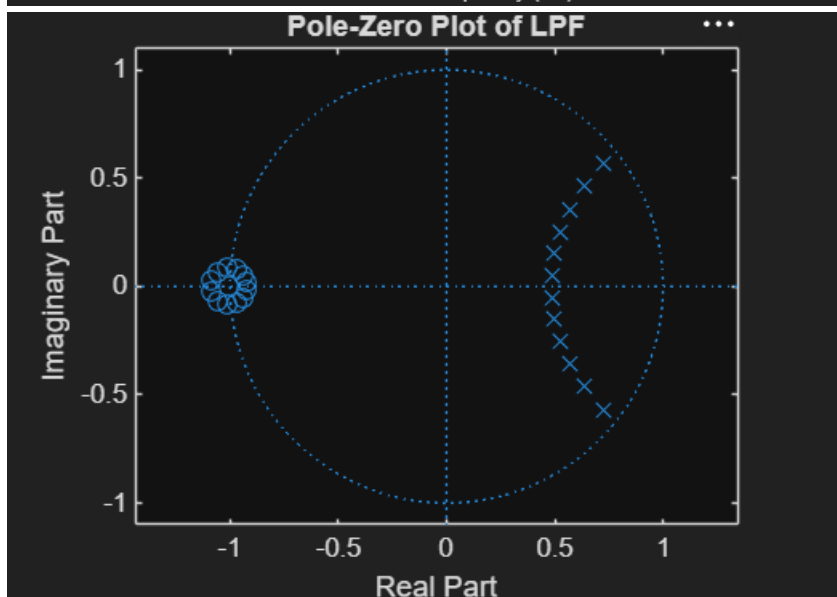
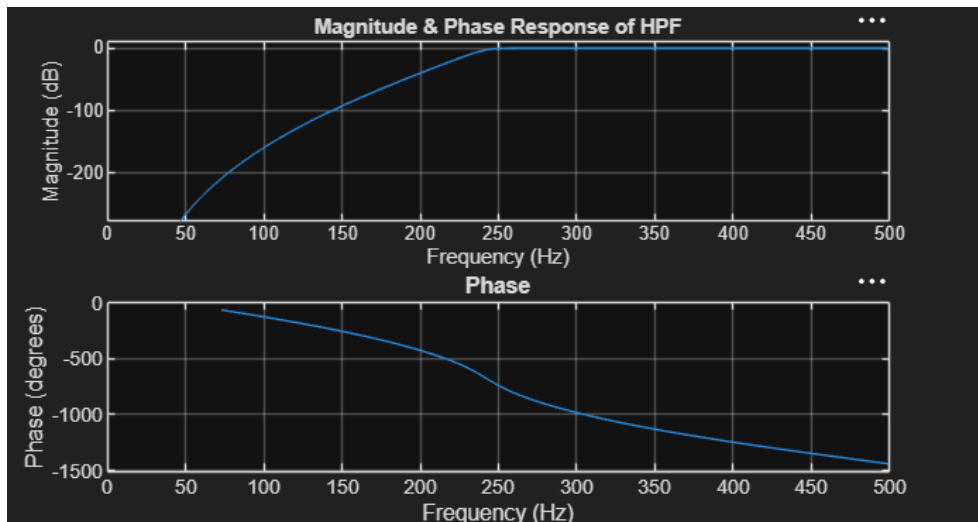


SIMULATION EXPERIMENT – 1

1.Design, Realization and Performance Analysis of Butterworth IIR Low-Pass and High-Pass Digital Filters Using MATLAB

OUTPUT:





RESHEKESH B (192512123)

***** LOW PASS FILTER *****

Filter Order = 12

Numerator Coefficients (b):

1.0e-03 *

0.0002 0.0027 0.0146 0.0488 0.1098 0.1757 0.2050 0.1757 0.1098 0.0488 0.0146 0.0027 0.0002

Denominator Coefficients (a):

1.0000 -6.8844 22.4341 -45.5100 63.7725 -64.8501 48.9643 -27.6092 11.5218 -3.4662 0.7128 -0.0899 0.0053

Cutoff Frequency = 106.3544 Hz

***** HIGH PASS FILTER *****

Filter Order = 17

Numerator Coefficients (b):

Columns 1 through 15

0.0001 -0.0013 0.0100 -0.0502 0.1756 -0.4566 0.9132 -1.4350 1.7938 -1.7938 1.4350 -0.9132 0.4566 -0.1756 0.0502

Columns 16 through 18

-0.0100 0.0013 -0.0001

Denominator Coefficients (a):

Columns 1 through 15

1.0000 -0.5247 2.4322 -1.0556 2.2461 -0.7954 1.0097 -0.2858 0.2357 -0.0517 0.0283 -0.0046 0.0016 -0.0002 0.0000

Columns 16 through 18

-0.0000 0.0000 -0.0000

Cutoff Frequency = 242.2731 Hz

LOW PASS FILTER

Passband Edge : 100 Hz

Stopband Edge : 150 Hz

Cutoff Frequency : 106.35 Hz

HIGH PASS FILTER

Stopband Edge : 200 Hz

Passband Edge : 250 Hz

Cutoff Frequency : 242.27 Hz

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